

Kavinda Weerasinghe

Software Engineer

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Profile

Software Engineer with **1.5 years** of hands-on industry experience specializing in full-stack development with **React, Node.js, and ASP.NET Core**, skilled in engineering **frontends, RESTful APIs, AI-powered services** and **microservices**-based enterprise applications deployed with **Docker, Kubernetes**, and **CI/CD** on **AWS/Azure** within a collaborative **Scrum/Agile** environment, while **leveraging AI** tools to accelerate delivery. **Ready** to learn, grow, and adapt while delivering impactful solutions.

Technical Skills

Languages	JavaScript, TypeScript, C#, Java, Python	CI/CD	Docker, Kubernetes, FluxCD
Frontend	React.js, Next.js, Redux, HTML, CSS, Material UI (MUI), Tailwind, Ant Design	Cloud	Azure, AWS
Backend	Node.js, Express.js, TypeORM, ASP.NET, EF Core, FastAPI, Spring Boot	AI Dev	LLM, RAG, Langchain, Azure OpenAI
Databases	PostgreSQL, MSSQL, MongoDB, Snowflake	AI Tools	Claude Code & Design, Codex
		Others	Prometheus, Grafana, Git, Puppeteer, Lens IDE, Figma, Firebase, Postman, WordPress

Professional Experience

Creative Software

May 2025 – May 2026

Software Engineer - Intern (12 months)

Intellectual property and brand protection system

- Shipped **5+ major features** on a **multi-tenant microservices** platform following **OOP, SOLID** principles and **layered architecture** within an **Agile/Scrum** using **React.js, Node.js, TypeScript, Spring Boot** and **PostgreSQL**.
- Developed **Puppeteer**-based scraper modules for **4+** social media and e-commerce platforms, achieving **95% accuracy** by engineering DOM extraction logic to overcome complex dynamic UIs and anti-bot detection mechanisms.
- **Reduced wasted S3 storage** and redundant **report regeneration** by building a full-stack report-management feature using React.js, Material UI, and Spring Boot REST APIs to let users retrieve, view, and delete past reports.
- **Decoupled the scraping pipeline** into separate result and item scrapers over **BullMQ message queues**, prioritizing concurrent listing collection with keyword searches and **eliminating Node.js thread blocking**.
- Designed and developed a full-stack POC with a **Next.js** frontend, **Material UI**, and **NestJS** data service to validate the feasibility of the verified seller and product identification feature.
- Spearheaded the **KYC customer onboarding** application by engineering a **configurable, template-driven** dynamic forms application using **React.js, Context API**, and **custom hooks**, and integrating **identity data providers** to the automated data fetching flow.
- Deployed and managed containerized microservices using **Docker** and **Kubernetes (EKS)** on, **CI/CD** automated with **AWS CodeBuild** and **Flux CD**, and monitored K8s cluster using **Lens IDE**.

Technologies used: TypeScript, C#, React.js, Next.js, ASP.NET Core, NestJS, Node.js, Express.js, PostgreSQL, TypeORM, Puppeteer, Java, Spring Boot, Docker, Kubernetes, AWS, Bullmq, MUI, FluxCD, Redis, Lens IDE, Prometheus, Grafana.

Verdentra

Jun 2024 – Dec 2024

Software Engineer - Intern (6 months)

Office Seating and Parking Reservation System

- Improved application usability by building an automated waitlist that reassigns freed slots, vehicle-plate display, and scheduled data-export jobs using **React.js, Node.js, Azure DevOps** and deployed on **Azure Functions**.

Employee Performance Tracking System

- Contributed to the application's UAT release by **resolving 20+** UI and business-logic issues and delivering a role-based annual employee-feedback feature using **React.js** and **ASP.NET Core** with API integrations.

Snowflake Migration Project

- Migrated **15+** reporting **stored procedures** from on-premises **MS SQL Server** to **Snowflake** Data Warehouse and refactored and optimized queries to improve query performance and reduce infrastructure costs.

Technologies used: React.js, TypeScript, Node.js, SQL, C#, ASP.NET Core, Azure, Ant Design, Snowflake

Open Source Contributions

OpenMRS - Draw on a Body Diagram Feature Development on O3 — Wiki 📖 — Discussions 💬

Jul 2026 - Present

- Implementing a feature for the OpenMRS O3 EMR platform for visualizing patient conditions through body diagrams to help doctors to explain war-injured patients' conditions more clearly. Conducted requirement gathering and business analysis with ICRC, UI Mockup design, and currently working on initial micro-frontend development

Technologies used: TypeScript, React, Carbon Design System, Claude Design

Education

Bachelor (Hons) of Information and Communication Technology in Software Technology University of Sri Jayewardenepura	GPA: 3.55 / 4.00 Second Upper (Upper Division)	Jul 2021 – Mar 2025
Diploma in Software Engineering Developers Stack Academy		2023 – 2024

Projects

Context Core - Organizational Context Engine for Software Product Companies (Individual) 	Aug 2026 - Present
Building a solution to unify scattered organizational knowledge (GitHub, Jira, Confluence, Slack, documents) into a single context layer with permission scopes that generates evidence-based, cited answers for engineers, other employees, and AI agents to accelerate onboarding and reduce repeated context-reconstruction time and token usage. Identified the problem, then designed and built the full-stack solution end-to-end, including the ingestion-to-retrieval pipeline. <i>Technologies used:</i> Python, TypeScript, FastAPI, React, MUI, LangChain, PostgreSQL, pgvector, Azure AI Foundry	
Your AI - Knowledge Graph, AI RAG for Domain Information Extraction with Glossary (Group) 	Aug 2024 - Apr 2025
Final-year research project building a domain-specific chatbot using an LLM and RAG pipeline over a knowledge graph constructed from PDF documents. Designed and implemented the React frontend and UI, and collaborated on the AI/RAG development. <i>Technologies used:</i> LangChain, Llama, MongoDB, Neo4J, Flask, Python, Typescript, React.js, Tailwind, Azure Services	
Lilac Store - Full-Stack E-Commerce Platform (Individual) 	Dec 2023 - Feb 2024
Built an end-to-end clothing e-commerce platform with customer-facing ordering, profiles, cart, and order history, plus an admin dashboard for inventory and sales insights. Implemented JWT authentication, image uploads, and Redux state management. <i>Technologies used:</i> JavaScript, React.js, Node.js, Express.js, MongoDB, Firebase, Tailwind, Redux	
Music Muse — Emotion-Based Music Recommender (Group) 	Oct 2023 - Feb 2024
Built the backend REST API and Spotify integration for a recommender that suggests music from a user's detected emotion, play history, and playlists; also contributed to the React frontend and the ML recommendation server. <i>Technologies used:</i> JavaScript, React.js, Node.js, Express.js, MongoDB, Firebase, Tailwind	
Dot Talk — Full-Stack Live Messaging & Communication Platform (Group) 	Sep 2023 - Mar 2024
Featuring a real-time chat application that facilitates real-time messaging, multimedia sharing, and group chats. <i>Technologies used:</i> JavaScript, React.js, Node.js, Express.js, MongoDB, MUI, Redux	

Publication

Knowledge Graph-Based Retrieval Augmented Generation System for Domain-Specific Information Extraction with Glossary-Aided Responses 
2026 6th International Conference on Advanced Research in Computing (ICARC)

Certifications

SQL (Basic/Intermediate) Certificate — HackerRank 	Feb – 2026
Kubernetes Basic Concepts and Configuration — Pluralsight 	Sep – 2025
API Security Practices — Pluralsight 	Oct – 2025
Generative AI with Large Language Models — Coursera 	Jan – 2025
Building APIs with C# and ASP.NET Core — Frontend Masters 	Dec – 2024
C# and .NET Basics — Frontend Masters 	Dec – 2024
Introduction to Microsoft Azure Cloud Services — Coursera 	Feb – 2024
Principles of UX/UI Design from Meta — Coursera 	Feb – 2024

Extracurricular

Host of KataKatha Communciation sessions — Creative Software	Dec 2025 – May 2026
Facilitated sessions focused on improving communication and presentation skills among peers.	
Tech Blog — Medium 	Sep 2025 – Present
Participant(4th Place) of VerdentraX Entrepreneurship Hackathon (2024) — Verdentra	2024
Competed among 6 teams to develop products and pitch innovative business ideas.	
Member of IEEE Student Branch — University of Sri Jayewardenepura 	Sep 2023 – May 2024
Volunteer as an Organizing Committee Member for J'Pura Expedition 2024	
Player in School Western Brass Band — Royal College, Horana	2014 – 2016

References

References available upon request.